<u>Statistics</u>
<u>Lecture 01</u>
<u>Basic theory of probability</u>

Terminology
- Sample space, outcomes, events
- Union, intersection, complementary, disjoint

Definition of probability

Axioms of P( ) operator

Addition rules
- Union of two, three, … events
- Disjoint set
- Complementary set

Conditional probability
- Definition and motivation in two ways
- Example
- Multiplication rule
- Law of total probability, example again
- Bayes theorem, example again

Independence

Combinatorics
- Permutations (ordered sequences)
- Combinations (unordered sequences)